Introduction to Artificial Intelligence

Why Is AI Suddenly Everywhere?

ECON6067: Quantitative Tools for Economic Analysis: From Math to AI
Lecture 1

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Today's Roadmap: One Storyline

We follow a single question all the way through:

How did one chatbot become the fastest-growing product in history — and is this time really different?

1 The Phenomenon

What does "going viral" really look like?

2 Four Waves of AI

Is this time genuinely different?

3 Why Now?

Why did deep learning wait until 2022?

4 Predict vs. Generate

How does next-token prediction create text?

5 Substitute or Complement

What does this mean for economists?

6 From Chat to Agents

Where this whole course is heading.

The Fastest Climb in History - ChatGPT

How long to reach 100 million users?

Since Nov 30, 2022 …

5 days

to reach the first 1 million users

2 months

to reach 100 million users

1 billion

monthly active users by May 2026

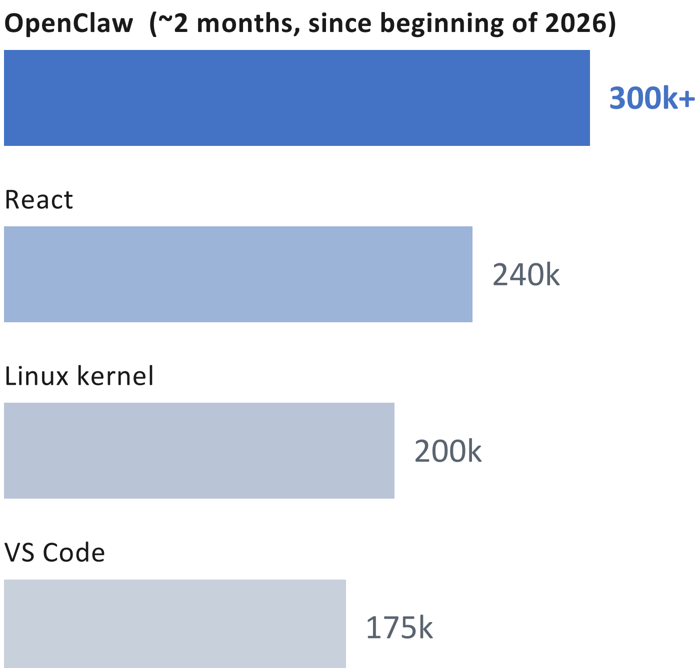
"In 20 years following the internet space, we cannot recall a faster ramp in a consumer internet app." — UBS

Source: Reuters / UBS

And the Next One Is an Agent

A second phenomenal product — OpenClaw ("the little crayfish"), an open-source AI agent

GitHub stars — fastest repo ever to the top



300k+ stars in ~2 months — it overtook the Linux kernel to top GitHub all-time.

Inside chat

lives in WeChat / Feishu / DingTalk / Telegram — 25+ channels, no app to open

It acts

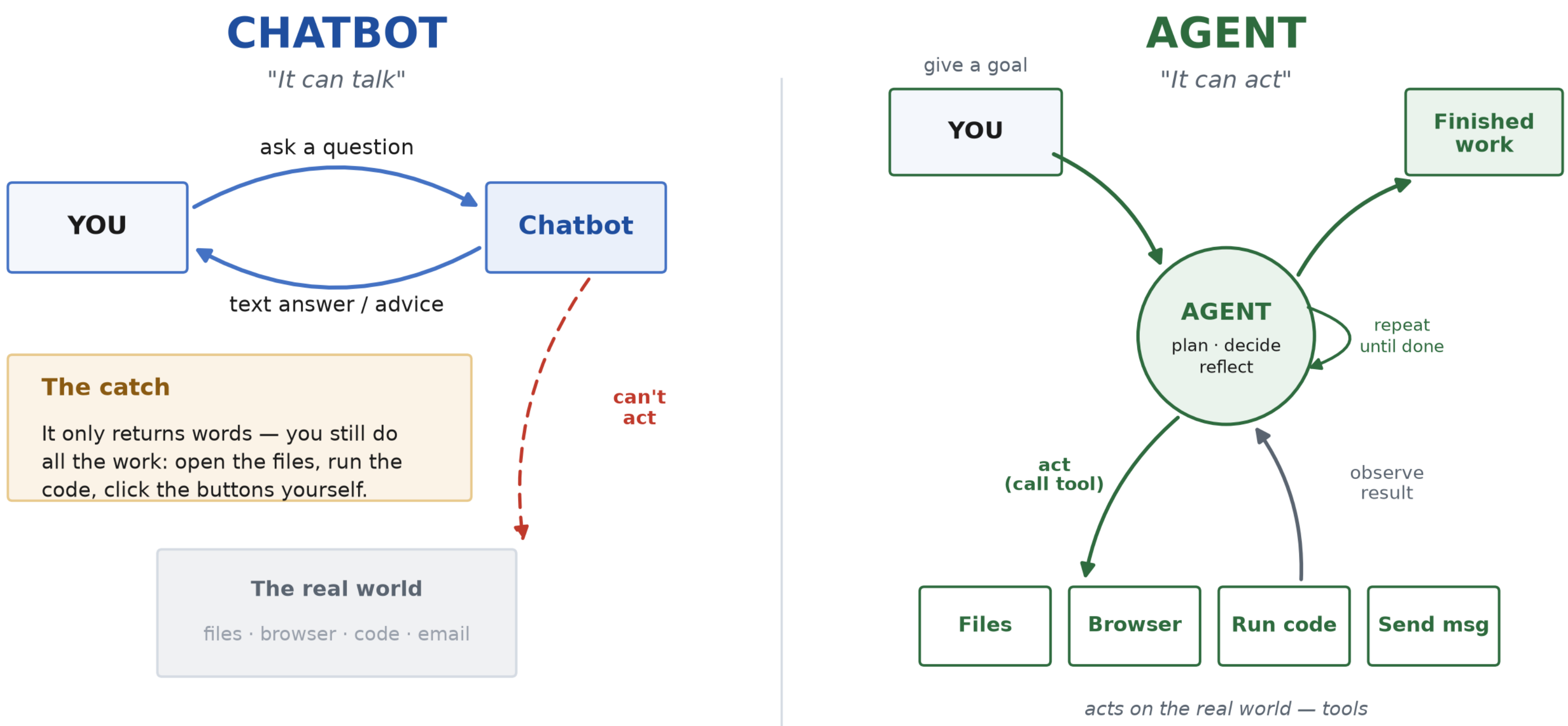
reads & writes files, runs commands, controls the browser, books flights, sends messages

700+ skills

community plugins + three-tier memory — the more you use it, the more it knows you

Source: Tencent Cloud, bakedwith.com

Chatbot vs. Agent



Chatbot = a mouth (it talks, you act). Agent = hands (you set the goal, it plans, uses tools, and delivers the finished work).

The tell: ChatGPT went viral as a **chat window**. The very next phenomenon is an **agent that does the work** — AI has moved from a "mouth" to "hands."

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How Tool Calls Work

The model proposes the action. The agent executes it.

The LLM does not run code directly. When it decides a tool is needed, it returns a structured message:

```
{
  "tool": "sql_query",
  "query": "SELECT ticker, close FROM prices WHERE date = '2024-12-31'"
}
```

The agent's execution layer reads that message, runs the actual tool, and sends the result back to the LLM. The LLM reads the result and decides what to do next.

An agent can use multiple tools in sequence — query a database, pass results to Python for analysis, generate a chart, and assemble everything into a PowerPoint deck — all from a single instruction.

The Agent Loop

01

Plan

The LLM reads your request and breaks it into steps.

02

Execute

The agent calls a tool — runs code, queries a database, writes a file.

03

Observe

The agent checks the result of the action.

04

Iterate

If unsatisfactory, the agent adjusts its plan and tries again.

When an agent encounters an error, it doesn't stop and ask you what to do. It diagnoses the problem, revises its approach, and tries again. This **autonomous problem-solving** is what separates agents from simple automation scripts.

Examples of Agent Tools

Python

Execute code, run analyses, generate plots.

Database

Query structured data with SQL.

Web Browser

Search the web and read live pages.

File System

Read, write, and organize files locally.

API Calls

Reach any external service via HTTP.

An agent may be connected to **multiple tools**. For example, get data from a database server, analyze it in Python, and post the summary through an API.

DEMO 1

LLM Alone vs. Agent with Tools

Same question. Two very different answers — because one has tools, one doesn't.

A · STUDENTS TRY

`chatgpt.hku.hk`

HKU ChatGPT — LLM only, no browsing

B · INSTRUCTOR DEMO

`perplexity.ai/computer`

Agent with tools — actually acts

~ 6 min

AI Is Not New

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History Timeline of AI

The term "Artificial Intelligence" was coined in 1956.

Era 1 | Symbolic AI

People set **Rules**, logic, and narrow reasoning.

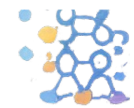
Era 2 | Expert Systems

Specialist knowledge was stored as ‘if–then’ rules.










1950	1956	1972	1980	2012	2016	2017	2022
Turing	Logic Theorist	MYCIN	XCON	AlexNet	AlphaGo	Transformer	ChatGPT

Era 3 | Statistical ML

Learning from data for OCR and scoring.

Era 4 | Deep Learning & Language Models

Text, images, and chat-native models.

Source: IBM: https://www.ibm.com/think/topics/history-of-artificial-intelligence

Era 1 | Symbolic AI and early demonstrations (1950s–1970s)

1950 | Turing
Frames AI as a question: can a machine behave intelligently?

1956 | Logic Theorist
Shows symbolic search can imitate theorem proving.

1966 | ELIZA
A warning that fluent language can create the illusion of understanding.

Core characteristics

- Knowledge was written as explicit rules and procedures.
- Search handled narrow, well-defined reasoning tasks.
- The logic was inspectable, so outputs felt transparent.

Alan Turing

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DEMO 2

Talk to ELIZA (1966) vs. ChatGPT

The first chatbot is 60 years old. Same idea — very different behavior.

OPEN THIS

`findingeliza.org`

Scan to try

~ 3 min

Era 2 | Expert systems and the first AI winters

Expert systems

Human expertise written as rules: if the facts look like X, ask Y, apply rule Z, and explain the conclusion.

Explicit rules worked best when expertise and exceptions could be written down clearly.

Two Examples:

1972 | MYCIN

A medical expert system that diagnosed infections and recommended antibiotics.

1980 | XCON

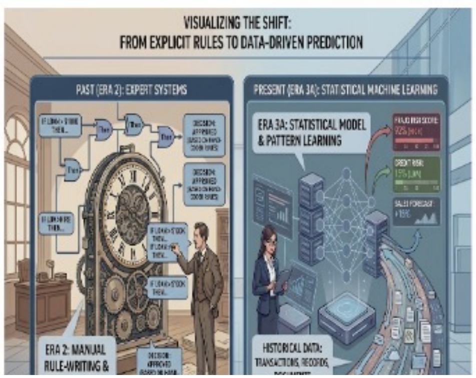
A production-rule system at DEC that configured VAX computer orders.

Why this wave stalled — and why that matters historically

Rules were expensive to build, hard to update, brittle when conditions changed, and weak when data were noisy or incomplete. That helps explain the later turn toward **machine learning**, where the system learns patterns instead of relying on every rule being hand-coded.

Era 3 | Statistical machine learning enters business workflows

Statistical ML learned patterns from **historical data** instead of relying on hand-coded rules.



The shift moved AI from explicit logic to data-driven prediction.

Various applications:

Credit scoring

Predict default risk from many historical examples.

OCR and document tagging

Turn messy documents into searchable, classifiable records.

Business adoption accelerated

- Better performance on noisy, high-volume cases
- Scalable screening, ranking, and prediction
- Practical gains in risk, operations, and reporting

The method changed, but the interface did not.

Most systems still depended on structured data, labels, features, and model pipelines rather than natural-language interaction.

This era made AI useful at business scale, but it still required specialists, curated data, and **engineered pipelines**.

Era 4A | Deep learning makes AI visible to the wider world

By the 2010s, neural networks delivered major gains in image, speech, and pattern recognition.

2012 | AlexNet

A breakthrough in image recognition

2016 | AlphaGo

A public milestone

AlphaGo made AI progress visible to the public.

Deep learning improved classification, perception, and extraction.

Business and research uses

Speech and call analysis
More usable transcription and speech analytics

Image-based checks
Inspection became easier to process automatically.

Limits at this stage

This era improved perception, not fluent **text collaboration**.

Deep learning expanded what AI could perceive, while natural-language interaction was still limited.

Era 1 | Symbolic AI and early demonstrations (1950s–1970s)

2017 | Transformer — "Attention Is All You Need"

A new neural-network architecture blueprint.

Core innovation: self-attention — each word can directly attend to every other word in the sequence.

2018 | BERT: The Encoder Path

Built on the Encoder half of Transformer.

2018–22 | GPT → GPT-3 → ChatGPT: The Decoder Path

Built on the Decoder half of Transformer.

Masters text GENERATION: chat, writing, coding, reasoning.
Scaling law: more data + bigger model

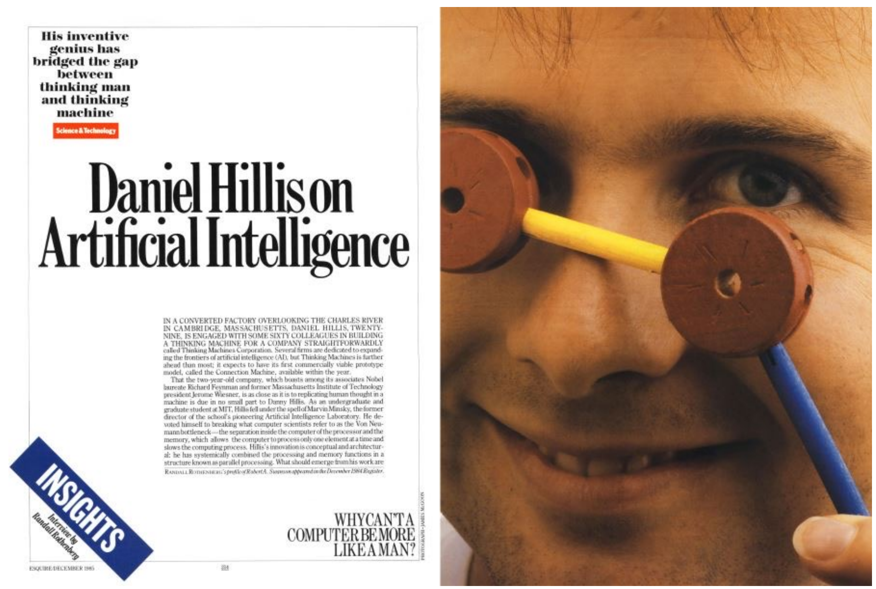
2024+ | The Decoder Path Dominates

ChatGPT, Claude, Gemini, DeepSeek — all decoder-based.

What changed technically and commercially

- models can read and write natural-language text **at scale**
- the user interface becomes a **prompt**, not a specialized tool
- retrieval lets models ground answers in documents
- falling cost + easy chat interface accelerate adoption

Is This Time Different?



Before AlexNet: What Was the ImageNet Challenge?

A benchmark competition (2010–2017) that let everyone measure progress on the same task, same data.

The setup

Task

Given a photo, name the object. 1,000 categories (dogs, kites, revolvers, kimchi...)

Data

1.2 million labeled training images, curated by Fei-Fei Li's team from ~15M ImageNet

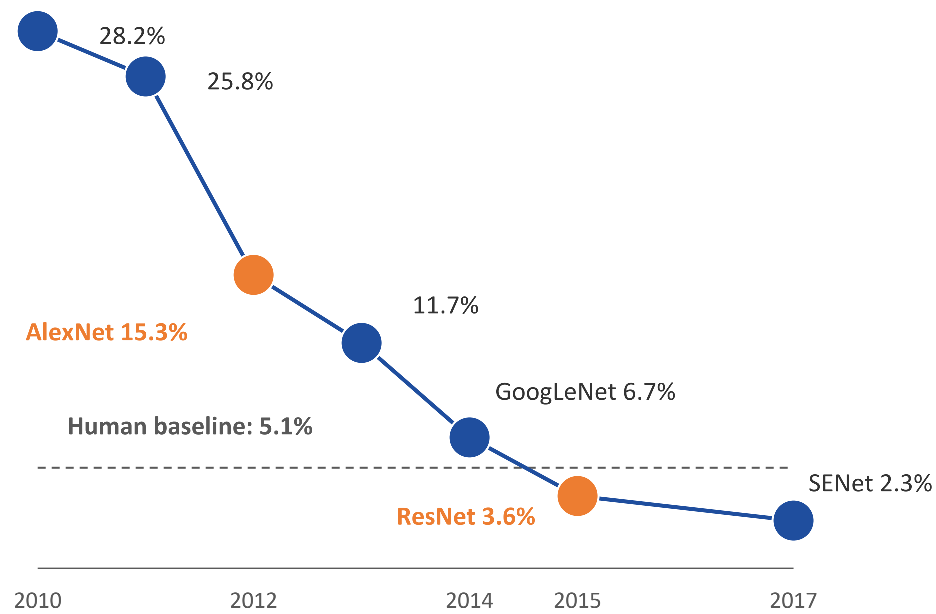
Metric

Top-5 error — model gives 5 guesses; wrong only if none match. Lets models hedge on similar categories.

Rules

Same training set, same test set, same day of announcement. A clean arena, every year.

Winning top-5 error, year by year



Fei-Fei Li's bet: better data > better algorithms.

5 years later, machines crossed the human line.

Source: Russakovsky et al. 2015 (ILSVRC); Karpathy 2014 (human baseline)

2012: AlexNet Made Deep Learning Impossible to Ignore

The number that changed the field

26.2%
Previous best
Top-5 error rate before AlexNet on ImageNet 2012

→

15.3%
AlexNet
Top-5 error rate from a deep CNN trained with GPUs

→

about 42%
Error reduction
A 10.9 percentage-point drop, not an incremental gain

The historical meaning: deep learning became credible.

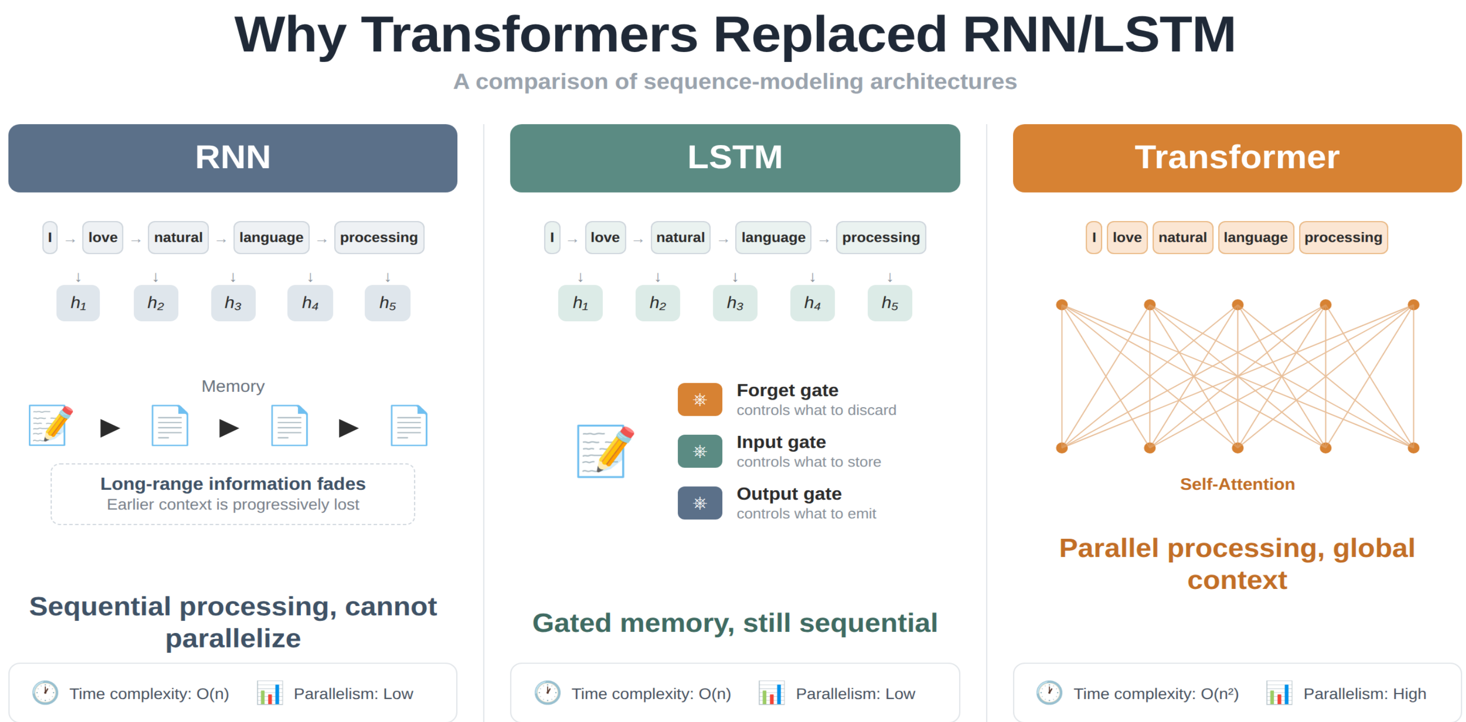
The **ImageNet** challenge created a clean arena: same dataset, same task, comparable results.

AlexNet moment did not yet give us ChatGPT. It showed something deeper: with enough data and compute, neural networks could learn useful representations rather than rely on hand-written rules.

"Ilya thought we should do it,
Alex made it work,
and I got the Nobel Prize."
— Geoffrey Hinton

Source: Krizhevsky et al. 2012; ImageNet / ILSVRC history

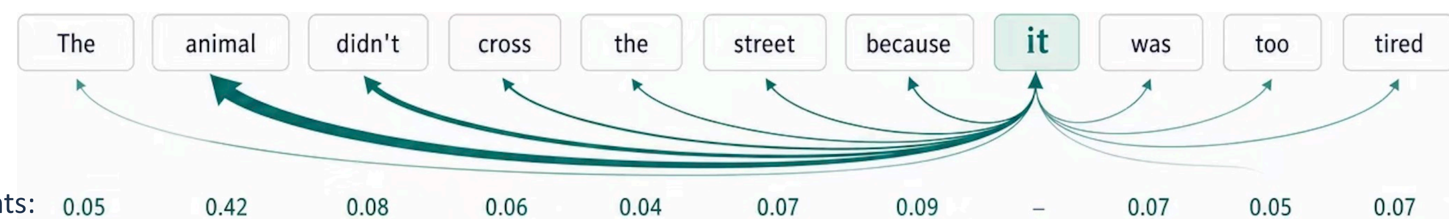
2017: “Attention is All Your Need” Transformer



Attention helps model connect words that are far apart, but meaningfully related.

Example:

Attention weights:



Source: Vaswani et al. 2017, Attention Is All You Need

One Transformer Family: BERT and GPT

Example: *The company reported strong revenue growth.*

BERT: fill in the missing token

Input:
The company reported strong [MASK] growth.

The model can look left and right:
strong + growth → revenue

So, BERT is often useful for understanding a sentence.

GPT: predict the next token

Input:
The company reported strong []

Next token: revenue

Then it predicts again: The company reported strong revenue → growth

Same Transformer idea, different training tasks

BERT learns meaning by recovering missing tokens from context. GPT learns generation by repeatedly predicting the next token. Both use attention to decide which tokens matter.

Source: Vaswani et al. (2017); Devlin et al. (2018); Radford et al. (2018)

What Count as a Token?

A **token** is the unit the model reads and generates; API usage is measured in tokens.

Rule of thumb for English

1 token ~ 4 characters
1 token ~ ¾ a word

Input and output

Prompt = input tokens.
Answer = output tokens.

Why it matters

Token counts affect context length, cost, speed, and how much text the model can handle at once.

Examples:

Text	My favorite color is red.
Token IDs	[3666, 4004, 3124, 318, 2266, 13]
	My favorite color is Red.
	[3666, 4004, 3124, 318, 2297, 13]
	我是香港大学经管学院的学生。
	[176389, 23485, 43065, 7697, 10052, 89469, 1616, 52770, 788]

Key point: models do not see words exactly as we do; they see token sequences.

Source: OpenAI Help Center, "What are tokens and how to count them?" (updated 2026)

2017: Why Transformer Changed Language AI

Key ideas

01
The problem
Older sequence models processed text step by step.
Slow and hard to scale.

02
Attention
Each token can look at other relevant tokens.
Context becomes easier.

03
Parallelism
Training can use modern GPUs more efficiently.
Compute scales better.

04
Scaling
More data + compute improves capability.
Architecture scales.

Transformer solved the language-scaling bottleneck. It did not make models intelligent by itself; it made it practical to train very large language models on very large text corpora.

A simple view:

TOKENS → ATTENTION → REPRESENTATIONS → NEXT TOKEN

Source: Vaswani et al. 2017, Attention Is All You Need

Once the algorithmic path was clear,

what was still missing?

- Data: enough high-quality text, code, and documents?
- Compute: enough affordable GPU-scale training and inference?
- Productisation: a chat interface ordinary people could use?
- …

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DEMO 3

See a Transformer Predict Live

The whole Transformer story on one page — running GPT-2 in front of you, no install.

OPEN THIS

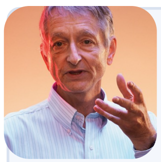
poloclub.github.io
/transformer-explainer

Scan to try

~ 4 min

Three Drivers Enter Together

ALGORITHM ▶ COMPUTE ▶ DATA ▶ INTERFACE



Geoffrey Hinton

Deep learning

AlexNet made neural networks feel practical at scale.

Jensen Huang

GPU compute

Massive parallel computation made training economically feasible.

Fei-Fei Li

ImageNet data

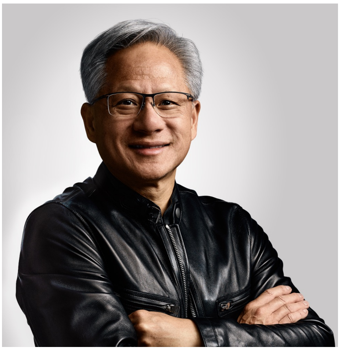
Large labeled datasets made progress measurable.

The point: none of the three drivers works alone

AlexNet and Transformer clarify the algorithm story. But the 2022 wave also required internet-scale data, GPU-scale compute, and a usable product interface. This is why the boom feels sudden even though the research history is long.

Source: Hinton personal page; NVIDIA; Stanford Profiles

Jensen Huang: Why GPUs Matter for AI

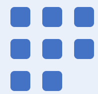


Jensen Huang, NVIDIA

A simple contrast

CPU: a few strong workers

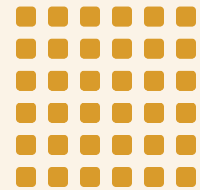
Good at complex, flexible tasks: branching, coordination, and operating-system work.



fewer powerful cores

GPU: many parallel workers

Good at doing the same kind of math many times at once. That is what neural networks need.



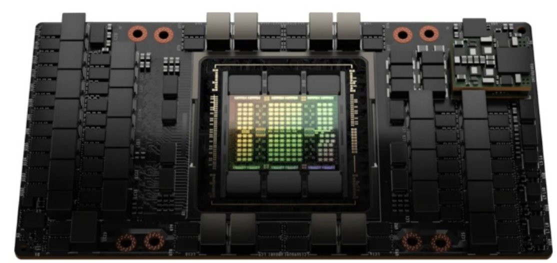
many parallel cores

NVIDIA did not start as an AI company; its gaming GPUs became essential to AI because both graphics and **neural networks** depend on massive parallel computation. So larger models become trainable and usable at lower time cost.

Jensen Huang's broader importance: NVIDIA made GPU computing a platform, not just a graphics component.

GeForce 256,
“The World’s First GPU”

The H100: The GPU That Powered the Generative AI Boom



Source: NVIDIA CUDA / GPU computing materials; course synthesis

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Fei-Fei Li: Data Made Vision Learnable

Fei-Fei Li, ImageNet

The problem was not only algorithmic

Raw data

Millions of images are not automatically useful.

A model needs examples connected to reliable labels.

Labeled data

ImageNet organized images into categories.

The labels turned pixels into training examples.

Why ImageNet changed the field

ImageNet did two things at once: it supplied large-scale labeled data, and it created a benchmark where researchers could compare progress. That made deep learning's improvement visible and measurable.

From ImageNet to Today: Data Became Strategic

- Frontier AI needs not just more data, but better data.
- The scarce resources are now quality, ownership, verification, and permission

If everyone has access to the same public internet, what data can still create an advantage?

Source: ImageNet / ILSVRC

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DEMO 4

Train an Image Classifier in 30 Seconds

You just heard about ImageNet. Now you become the labeler.

OPEN THIS

```
teachablemachine
.withgoogle.com
```

Scan to try

~ 4 min

Why Now? The Joint Effect

The boom came from the interaction, not one isolated invention:

DATA × COMPUTE × ALGORITHM ▶ From Single Tasks to General Capability ▶ CHAT → AGENT ▶ GENERAL-PURPOSE TECH

Algorithm

AlexNet -> Transformer

Deep learning became scalable for perception and language.

Compute

GPU ecosystems

Training and serving large models became economically feasible.

Data

Internet-scale corpora

Text, code, and documents became training material.

Then productisation made the capability visible

ChatGPT did not invent the whole stack in 2022. It packaged a long research trajectory into a simple interface. That is why this wave felt sudden to the public.

Source: Krizhevsky et al. 2012; Vaswani et al. 2017; Brown et al. 2020

From Why Now to What It Does

We have explained why the AI wave became possible.

Now we need the operating logic:

How can repeated prediction produce writing, reasoning, and code?

1. Familiar prediction

Predict a price, a risk score, demand, or return.

→

2. Sequential prediction

Predict the next token, append it, then predict again.

→

3. Useful output

Many small predictions become an answer, a paragraph, a plan, or code.

Prediction is familiar. The surprise is what prediction becomes when it is repeated at scale with huge data, compute, and Transformer models.

Two Tasks That Look Worlds Apart

TASK 1

Predict a house price

$ 7,420,000

Given size, location, age → output one number.

A task economists know well: regression.

TASK 2

Write a paragraph

"The housing market in Hong Kong reflects..."

Given a prompt → output fluent, original text.

A task that feels almost human.

Same underlying machinery? Or two different kinds of prediction?

If It Generates, Why Call It Prediction?

The puzzle

The surprising answer: generative AI is built from repeated prediction.

THE FAMILIAR

Regression / forecasting

Fit parameters to data, minimise error, predict a number or a class. Linear models, trees, neural nets.

THE NEW

Generative models

Produce new text, images, code — outputs that never appeared verbatim in the data.

ChatGPT seems to create new text, but its basic training task is simple: given the text so far, predict what should come next.

Now connect it to language: generation is prediction repeated token by token.

Can predicting the next token

really produce writing, reasoning, and coding?

Consider these angles:

- What does the model need to know to choose the next word?
- Does one correct next word always exist?
- What happens if it repeats this process thousands of times?

Token-by-Token Prediction

Key idea

TEXT SO FAR

Hong Kong housing prices are high because …

The model treats the prompt as context and asks: what token should come next?

This is the prediction setup.

NEXT TOKEN CHOICES

land / supply / demand / interest / …

Each choice has a probability. The model selects one token and appends it to the text.

Then the new text becomes the context for the next prediction.

Generation is prediction repeated

The model does not write the whole paragraph at once. It predicts a next token, adds it to the sequence, and repeats. A chain of small predictions becomes a sentence, then a paragraph, then an answer.

Discriminative vs. Generative — in Plain Words

Key idea

DISCRIMINATIVE

Draws the line

Learns to tell things apart: spam vs. not, default vs. repay, this price vs. that.

Answers "which / how much?"

The economist's regression lives here.

GENERATIVE

Draws the thing

Learns the shape of the data well enough to make new examples: a sentence, an image, a line of code.

Answers "what comes next?"

ChatGPT predicts the next token, over and over.

The key idea

Generative AI is not the opposite of prediction. It is sequential prediction at scale. The surprise is that a simple objective, when combined with huge data, compute, and model size, produces rich behavior.

Source: Goodfellow et al., Deep Learning, Wei et al. 2022, Emergent Abilities

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DEMO 5

How Chinese and English Get Tokenized

Tokens are the unit of billing and context. Chinese pays a hidden tax.

Scan to try

OPEN THIS

`platform.openai.com/tokenizer`

~ 3 min

From a Tool That Talks to One That Acts

Where this leads

The same chain now points at your own work.

CHAT

It can talk

Summarise a filing, explain a concept, draft a paragraph — you still do the work.

▶

AGENT

It can act

Given a goal, it plans steps, calls tools, and completes a whole workflow — then reports back.

Concrete example

- AI can read a company's 10-K filing and extract useful signals about risk and performance.
- A chat model helps you read faster.
- An agent can pull the filings, run the analysis, and hand you a draft — a step toward automating knowledge work itself.

If the Agent Can Finish the Job…

The economic question

…then what is a trained analyst, researcher, or economist worth?

SUBSTITUTE

AI replaces the worker

If an agent does the whole task faster and cheaper, demand for that labor falls — and so do wages.

COMPLEMENT

AI amplifies the worker

If AI handles the routine, skilled people do more of the high-value judgement — and become more valuable.

The economic question. Whether AI substitutes for or complements a worker is not fixed — it depends on the task, the skill, and on whether you can **direct** the AI rather than compete with it. That choice is partly yours to make.

Source: Acemoglu & Restrepo 2019, Autor 2015

Does AI lower or raise the value of someone who knows how to use it well?

And what does that mean for productivity and inequality?

Consider these angles:

- Whose wages rise — those who direct AI, or those who compete with it?
- If routine work gets cheap, where does human value concentrate?
- Could the same tool widen the gap between workers?

DEMO 6

Have an Agent Trace an Event Across Markets

The professional payoff — this is what the rest of the course teaches you to build.

The Fed started raising rates in March 2022. Show me how three assets moved over the next 12 months — the US 10-year Treasury yield, the Russell 2000, and Bitcoin. Make one comparison chart

Demo

`www.perplexity.ai/computer`

~ 5 min

The Course Contract

End-to-end agents give you capability. This course aims to give you more controllability

01 Can you open the black box?

From using an agent's answer to reading every step it took.

→ Lecture 2–3 · Vibe coding, transparent workflows

02 Can you make it your default, not the vendor's?

Package your data, your judgement, your style into skills you own.

→ Lecture 4 · CLAUDE.md, custom skills, reproducibility

03 Can you sign your name to the final deliverable?

From a 90-point black-box demo to a 95-point white-box you can defend.

→ Lecture 5 + Capstone · Multi-agent orchestration, sign off

Your Position — and Where This Course Goes

Takeaway

AI does well

First-pass reading · compression · pattern-finding · tireless scale

Humans do well

Judgement · asking the right question · challenging the output · deciding

And. Taking the responsibility!

WHERE THIS COURSE IS HEADING

Rather than be replaced by an agent, learn to build and direct one. The rest of this course teaches you to assemble agents in professional frameworks to automate your own workflows.

The economist's takeaway. A general-purpose technology raises total factor productivity, but the gains flow to those who learn to **complement** it. The skill you build here — directing agents — is how you stay on the complement side of the line.

Summary: One Chain, Start to Finish

Takeaway

DATA × COMPUTE × ALGORITHM → CAPABILITY LEAP → CHAT → AGENT → GENERAL-PURPOSE TECH

- The phenomenon was real: ChatGPT was the fastest-adopted product in history.
- AI is old and has frozen over twice.
- The real driver is the resonance of data × compute × algorithm crossing a critical point.
- Capability, natural-language access and ease of use are results of that driver, not separate causes.
- Interaction is evolving from chat (it talks) to agents (it acts) — automating whole workflows.

Take this away: this time is different because AI became a general-purpose technology — and because interaction is moving from chat to agents that act.

Next lecture: from idea to working agent — the building blocks you will use to automate your own workflows.

References & Further Reading

The basic concept of ML and DL

- An Introduction to Statistical Learning, with applications in R/ Python. By James, Witten, Hastie, and Tibshirani: https://www.statlearning.com/

The basic concept of LLM and how to use it

- Zhao W X, Zhou K, Li J, et al. A survey of large language models[J]. arXiv preprint arXiv:2303.18223, 2023, 1(2): 1-124.

The architecture of Transformer

- Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need[J]. Advances in neural information processing systems, 2017, 30.

Some Useful Videos

Machine Learning

- Introduction to ML, Intuitive and simple (7 min) https://www.youtube.com/watch?v=ukzFI9rgwfU
- Some basic concepts of ML (12 min) https://www.youtube.com/watch?v=Gv9_4yMHFhI
- All Machine Learning algorithms explained (17 min) https://www.youtube.com/watch?v=E0Hmnixke2g

Deep Learning

- Introduction to NN(DL) (5 min) https://www.youtube.com/watch?v=6M5VXKLf4D4
- Concepts of DL, series courses of DL , first course is basic concept (each video/ 17min) https://www.youtube.com/watch?v=aircAruvnKk

Large Language Model

- How LLM works and generates (7 min) https://www.youtube.com/watch?v=LPZh9BOjkQs
- Detailed introduction of LLM (1h) https://www.youtube.com/watch?v=zjkBMFhNj_g

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Glossary: How Models Work

Keep these terms handy when reading later slides or using AI tools.

Token

A small unit of text that a model reads or predicts, such as a word, part of a word, number, or punctuation mark.

Example: “valuation” may be one token, while “unbelievable” may split into parts.

Prediction

The model estimates what is most likely to come next, based on patterns learned from data.

In language models, this usually means predicting the next token.

Neural Network

A model made of connected layers that learn patterns from examples rather than relying only on hand-written rules.

It adjusts internal weights during training.

Deep Learning

A family of neural-network methods with many layers, powerful enough to learn complex patterns in images, text, and data.

AlexNet showed deep learning could work at scale for vision.

Transformer

A neural-network architecture designed to process tokens in context and support large-scale language models.

BERT and GPT are both built from Transformer ideas.

Attention

A mechanism that lets a model weigh which tokens matter most for understanding or generating a particular token.

It helps the model use context rather than read every word equally.

Glossary: Models, Data, and Compute

These terms connect the technology story to the three drivers behind the current AI wave.

BERT

A Transformer model trained to understand text by recovering missing tokens from surrounding context.

Useful for classification, search, and text understanding tasks.

GPT

A Transformer model trained to generate text by predicting the next token repeatedly.

Repeated prediction can produce paragraphs, code, plans, and answers.

Training Data

Examples used to teach a model patterns, such as labeled images, text, code, documents, or feedback.

ImageNet turned images into labeled training examples.

GPU

A chip built for parallel computation. It is useful for AI because neural networks require enormous repeated math.

Originally important for graphics and gaming; now central to AI.

Compute

The processing power, chips, memory, and systems needed to train and run AI models.

More compute makes larger models possible, but also more expensive.

Agent

An AI system that can use tools and take steps toward a goal, rather than only producing a chat response.

The course moves from chat to agents that can automate workflows.